

Exam 23/6/25

Question #1.a

$$L1 = (1, 2, 3, 8, 9, 15, 20, 35, 40)$$

$$L2 = (2, \textcircled{5}, 35)$$

$$L11 = (1, 2, 3)$$

$$L21 = (\textcircled{2})$$

match and return "2". The recursive calls then stop since $L2$'s calls are empty.

$$L12 = (8, 9, 15, 20, 35, 40)$$

$$L22 = (\textcircled{35})$$

match and return "35", and then stop as above.

Question #1.b

The blocks have size 3, and they are built for the longer list that is $L1$.

$$L1 = \boxed{1, 2, 3} \boxed{8, 9, 15} \boxed{20, 35, 40} \Rightarrow L1' = (1, 8, 20)$$

$$L2 = 2, 5, 35$$

We "merge" $L1'$ with $L2$ and thus get $\overline{1}, 2, \overline{5}, \overline{8}, 20, 35$ where we used a bar to denote elements of $L1'$.

We then decide that block 2nd of $L1$ is skipped and we intersect

$$(1, 2, 3) \cap (2, 5) \rightarrow 2$$

$$(20, 35, 40) \cap (35) \rightarrow 35$$

Question #2.1

We decompress the entire sequence of Elias Fano

$$L = 01\ 11\ 00\ 01\ 01\ 00\ 11\ 11\ 00\ 11\ 00$$

$$H = \frac{110}{0} \quad \frac{110}{1} \quad \frac{10}{2} \quad \frac{0}{3} \quad \frac{10}{4} \quad \frac{10}{5} \quad \frac{10}{6} \quad \frac{110}{7} \quad \frac{00}{8} \quad \frac{0}{9} \quad \frac{10}{10} \quad \frac{00000}{11} \dots$$

~~8~~ $h = 4 \text{ bits} \rightarrow \text{so } b = l + h = 5 \text{ bits}$

Notice that $|L| = 11$ and $|L| = 22$, so $l = 2$.

$$S = (\underbrace{0000\ 01}_1, \underbrace{0000\ 11}_3, \underbrace{0001\ 00}_4, \underbrace{0001\ 01}_5, \underbrace{0010\ 01}_9, \underbrace{0100\ 00}_{16}, \underbrace{0101\ 11}_{23}, \\ \underbrace{0110\ 11}_{27}, \underbrace{0111\ 00}_{28}, \underbrace{0111\ 11}_{31}, \underbrace{1010\ 00}_{40})$$

Question #2.2

Access (8) = correspond to the 8th bit equal to 1 in H, which is in the bucket 6 [this can be found thanks to $\text{Select}_1(8) - 8 = 14 - 8 = 6$] that we represent with 4 bits $\Rightarrow 0110$
We finally retrieve the 8th pair of bits in $L \Rightarrow (11)_2$

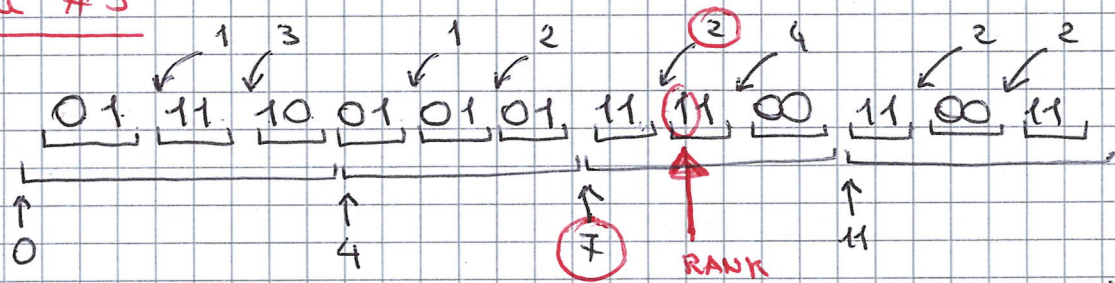
$$\text{Return } (011011)_2 = 27$$

Question #2.3

Next GFA (31) $\Rightarrow$ We represent 31 in 6 bits = $(\underbrace{011111}_h \underbrace{1}_l)_2$

the right part is $(011)_2 = 7$, so we access the 7th bucket by executing $\text{Select}_0(7) = 15$. Since $15 - 7 = 8$, we access the 9th bit equal to 1 $\Rightarrow$ i.e. it is the one storing the bucket 7, whose low part is $(00)_2$ and thus (28) . Then we access the next element, that is (31) , and stop.

Question #3



local rank for the 8 small blocks are shown above.
~~they are stored in the table of~~

T	0	1	
00	0	0	
01	0	1	
10	1	1	
11	1	2	

Rank (15) = 3rd superblock +
 2nd block +
 first position

$$= 7 + 2 + 1 = 10$$

Question #4

We have a binary tree encoded in $B[1, 2n]$ for its structure and $L[1, n]$ for its node labels.

We have to design an algorithm that counts "how many times a node labeled A has grandparent labeled B".

count = 0;

for $i = 4$ To $2n + 1$: // start from nodes that possibly have a grandparent

if $B[i] == 1$ then // node exists, no dummy

id1 = rank₁(B, i)

parent = $\lfloor \text{select}_1(\text{id1}) / 2 \rfloor$ // this is "i"

id2 = $\lfloor \text{select}_1(\text{parent}) / 2 \rfloor$

if $(L[\text{id1}] == "A" \wedge L[\text{id2}] == "B")$ count++;

return count;

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